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1a .You have 3 unexcused absences for lab.

1b. "USC campus support indicates an emergency has occurred"

1c. True

1d. I agree with the aspect that AI can act as a crutch. Especially if you just nod along with what it gives you, or do not necessarily understand what it is giving you. And the process of making a solution gives you insights into how to structure and break apart big problems. But I do not necessarily agree that asking ai about anything will instantly steal knowledge from you. From my own past experience, I do not believe that learning requires hours upon hours of failure and staring at code. It just requires a proper attempt at identifying and fixing it on your own. Afterwards, even if you do ask ai for help, you can properly internalize why.

As for how I will approach problems, especially when code does not work, It will most likely be the same as I have always done it. Stare at the problem, and reread the lines of code until I build a mental model of it, break down the problems, and simulate everything running one by one, and then once a block is identified, model what different solutions would do, and which would get me the fix I desire.

1e. False

1f. False

2a. Can lead to errors

2b. Can lead to errors

2c. Is fine

2c. git add

2e. git log -3

2f. git clone [git@github.com:usc-csci104-spring2026/resources.git](https://github.com/usc-csci104-spring2026/resources.git)

$$\begin{aligned}
 1. \quad 2 \cdot 2 &= 4 = 2^{2^1} \\
 4 \cdot 4 &= 16 = 2^{2^2} \\
 16 \cdot 16 &= 256 = 2^{2^3} \\
 \sim \sim &= 2^{2^k}
 \end{aligned}$$

Let's assume 2^{2^k} is more than n , and that i will reach 2^{2^k} .

Set $=$:

$$2^{2^k} = n$$

$$\log_2(2^{2^k}) = \log_2(n)$$

$$\log_2(2^k) = \log_2(\log_2(n))$$

$$k = \log_2(\log_2(n))$$

$$\Theta(\log \log n)$$

Problem b:

$$\text{Outer loop: } \sum_{i=1}^n \Theta(1) = \Theta(n)$$

Inner if condition:

Assume: $n = 9$

Dummy variable: T

Iteration it runs:

$$T = 1$$

$$T = 2$$

$$T = 3$$

what caused it

$$3 = 1 \cdot \sqrt{9}$$

$$6 = 2 \cdot \sqrt{9}$$

$$9 = 3 \cdot \sqrt{9}$$

$$\text{Generalized} = T \cdot \sqrt{n}$$

Dummy variable p :

$$p = 1$$

3^3 times

$$p = 2$$

$$6^3$$

$$p = 3$$

$$9^3$$

$$\text{Generalized: } (T \cdot \sqrt{n})^3$$

$$\sum_{T=1}^{\sqrt{n}} (T \cdot \sqrt{n})^3 \xrightarrow{\text{General Form}} n^{3/2} \cdot \Theta(\sqrt{n}^4) \rightarrow \Theta(n^{7/2})$$

$$\Theta(n^{7/2}) + O(n) = \Theta(n^{7/2})$$

Problem C:

$$O(n^2)$$

$$\sum_{i=1}^n$$

$$\sum_{k=1}^n$$

$$\sum_{m=1}^{\log_2 n}$$

$$O(n)$$



$$O(n) \cdot O(n) = O(n^2)$$

$$\sum_{m=1}^{\log_2 n}$$

$$A(k) = i$$

Each position k can only be correct one time per outer loop. In total the max work for this if would be n :

$$n \cdot \log_2(n)$$

$$n^2 + n \log(n)$$

$$n^2 >$$

$$\Theta(n^2)$$

Problem di

0	1	4	9	16	25	36	49	64	81
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$$30/2 = 15$$



$$T(n) = \sum_{i=0}^{n-1} O(1) = O(n)$$

Dummy k	If allowed	second loop
1	10	10
2	$10 \cdot \frac{3}{2}$	$10 \cdot \frac{3}{2}$
3	$10 \cdot \left(\frac{3}{2}\right)^2$	$10 \cdot \left(\frac{3}{2}\right)^2$
4	$10 \cdot \left(\frac{3}{2}\right)^3$	$10 \cdot \left(\frac{3}{2}\right)^3$

$$n \approx 10 \cdot \left(\frac{3}{2}\right)^{k-1}$$

$$\frac{n}{10} \approx \left(\frac{3}{2}\right)^k$$

$$\log_{\frac{3}{2}} \left(\frac{n}{10}\right) \approx k$$

$$T(n) = \sum_{k=1}^{\log_{\frac{3}{2}} \left(\frac{n}{10}\right)} 10 \cdot \left(\frac{3}{2}\right)^{k-1}$$

$$10 \sum_{k=1}^{\sim} \left(\frac{3}{2}\right)^{k-1}$$

$$1 = k-1$$

$$\sum_i^{\log_{1.5}(n)-1} \frac{3}{2}^i$$

$$O\left(\frac{3}{2}^{\log_{1.5}(n)-1}\right)$$

$$O\left(\frac{\frac{3}{2}^{\log_{1.5}(n)}}{\frac{3}{2}^1}\right)$$

$$O\left(\frac{n}{\frac{3}{2}}\right)$$

$$T(n) = \Theta(n) + O(n) = \Theta(n)$$

Problem 4a:

$$1, 2, 3, 4$$

2 - in 2 = 1, 2, 3, 4
1 - in 2 = 5, 6, 7

Trace Work answer
↓

Run 0: $llrec(1, 5)$ $1 \rightarrow 5 \rightarrow 2 \rightarrow 6 \rightarrow 3 \rightarrow 4$

Run 1: $llrec(5, 2)$ $5 \rightarrow 2 \rightarrow 6 \rightarrow 3 \rightarrow 4$

↓

Run 2: $llrec(2, 6)$ $= 2 \rightarrow 6 \rightarrow 3 \rightarrow 4$

↓

Run 3: $llrec(6, 3)$ $= 6 \rightarrow 3 \rightarrow 4$

↓

Run 4: $llrec(3, \text{nullptr})$ $= 3 \rightarrow 4$

Run 5: 3 $\rightarrow$ 4

Problem 4c:

int 1: nullptr

int 2: 2

Run 2: $llrec(\text{nullptr}, 2)$

↓

2

